Problem-Solution Fit

Date	27 June 2026
Team ID	-
Project Name	Smart Lender
Maximum Marks	5 Marks

PROBLEM-SOLUTION FIT

1. CUSTOMER SEGMENT(S) [CS]

- Banks
- Financial Institutions
- Credit Officers
- Loan Processing Teams
- Financial Analysts
- Individuals applying for loans

2. PROBLEMS / PAINS + ITS FREQUENCY [PR]

- Manual loan approval process is slow and time-consuming.
- Human errors may lead to incorrect loan decisions.
- Difficulty in identifying high-risk applicants.
- Large number of applications during peak periods increases workload.
- High risk of loan defaults and non-performing assets (NPAs).

3. TRIGGERS TO ACT [TR]

- Increase in loan applications.
- Need for faster loan approval.
- Requirement to reduce loan default risk.
- Demand for accurate and data-driven decision making.
- Digital transformation in banking services.

4. EMOTIONS BEFORE / AFTER [EM]

Before

- Stress due to lengthy approval process.
- Uncertainty about loan eligibility.
- Heavy workload for credit officers.
- Delays in customer service.

After

- Faster loan approval.
- Increased confidence in decisions.
- Reduced workload.
- Improved customer satisfaction.
- Better operational efficiency.

5. AVAILABLE SOLUTIONS (PROS & CONS) [AS]

Manual Verification

Pros:

- Human judgment.
- Flexible decision-making.

Cons:

- Slow process.
- Error-prone.
- Difficult to scale.

Traditional Rule-Based Systems

Pros:

- Easy to implement.
- Consistent rules.

Cons:

- Cannot adapt to changing financial patterns.
- Lower prediction accuracy.

6. CUSTOMER LIMITATIONS [CL]

- Limited historical credit information.
- Budget constraints for advanced software.
- Large number of loan applications.
- Time limitations.
- Data quality issues.
- Different applicant profiles.

7. BEHAVIOR + ITS INTENSITY [BE]

- Customers expect quick loan approval.
- Banks process hundreds of applications daily.
- Credit officers manually verify applicant information.
- Financial analysts require accurate risk prediction.
- High dependence on historical financial data.

8. CHANNELS OF BEHAVIOR [CH]

Online

- Banking websites
- Loan application portals
- Mobile banking applications
- Internet banking

Offline

- Bank branches
- Financial service centers
- Loan consultants
- Customer support offices

9. PROBLEM ROOT / CAUSE [RC]

- Manual evaluation consumes significant time.
- Lack of intelligent decision-support systems.
- Increasing volume of loan applications.
- Inconsistent evaluation methods.
- Inability to accurately predict applicant creditworthiness.

9. PROBLEM ROOT / CAUSE [RC]

Smart Lender is a machine learning-powered web application that predicts the creditworthiness of loan applicants using Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and XGBoost algorithms. After evaluating all models, XGBoost is selected as the best-performing model for real-time loan approval prediction. The Flask-based application enables banks and financial institutions to make faster, accurate, and data-driven lending decisions while reducing loan default risk and improving operational efficiency.